1

$\omega(G)$ is the size of the largest clique in G.

Suppose $\omega(G) \leq k$, then G would not satisfy the initial condition of having a $k+1$-clique thus $\omega(G) > k$

Suppose $\omega(G) > k+1$, then there exist at least $k+2$ vertices whose neighbours form a $k+1$ clique. By definition $G - v$ where v is one of these vertices would be a k-tree because it has $k+1$ clique but then $G - v$ would not be a k-tree because the neighbours of v form a $k+1$ clique instead of a k-clique. Thus $\omega(G) = k+1$

2

By definition if $H = G - v$ is a k-tree and $G = H + v$ is a k-tree iff $N(v)$ from a k-clique. Thus we know that we can disconnect v from G by removing the k neighbours of v. $H = G - v$ is not a k-tree then G must be K_{k+1} where $\kappa(K_{k+1}) = k$. Thus $\kappa(G) \leq k$.

For proving $\kappa(G) \geq k$ we can use induction.

Base case is $G = K_{k+1}$ which is known that $\kappa(G) = k$

Inductive step assume G is a k-tree and $\kappa(G) = k$ show $G{+}v$ is either a k-tree with $\kappa(G{+}v) = k$

When we add a new vertex v to a graph G then $\kappa(G + v) \geq \min(N(v), \kappa(G))$. If $G + v$ is a k-tree then $N(v) = k$ thus $\kappa(G + v) \geq k$.

3

Find vertices of degree k (these are the $+v$ vertices basically), the neighbours of these vertices form a clique

No two vertices of degree k are adjacent to eachother unless we are at K_n in which case we pick 1 and return 1 (or a forest of cliques in which case same idea of a vertex per clique)

For each vertex of degree k mark it as part of our independent set I then remove our marked nodes and their adjacent vertices

Repeat on the subgraph looking for vertices whose neighbours form a clique (not necessarily degree k) until the base case of a forest of cliques and pick one from each clique

Proof of correctness

Our independent set I is independent, because whenever we pick a vertex we immediately delete all of its neighbours, so nothing picked later can be adjacent to it.

Now suppose the algorithm picks $s_1, \ldots, s_m$, so $|I| = m$. Let Q_i be the set of vertices removed because of s_i (s_i plus whatever neighbours of it were still around and not already removed). Each Q_i is a subset of s_i's closed neighbourhood, which is a clique since s_i was simplicial, so each Q_i is a clique. Every vertex gets removed exactly once, so $Q_1, \ldots, Q_m$ split up all of $V(G)$. Any independent set can only contain 1 vertex from each clique, so any independent set has at most m vertices. Thus I is a largest independent set.

Each round we check each vertex's neighbourhood for being a clique in $O(n^2)$, so $O(n^3)$ per round, and there are at most n rounds, so $O(n^4)$ total, which is polynomial in $|G|$.

4

Lets call our subgraph of a k-tree H and the k-tree that H is a subgraph of as T. Any subgraph of H (only removing vertices and edges) would therefore be a subgraph of T because a subgraph of a subgraph of our k-tree would be a subgraph of the k-tree. In other words $G \subseteq H \subseteq T$

Thus the main thing we need to show is that contractions of H are still subgraphs of a k-tree.

For this we can do a proof by contradiction, suppose H is a subgraph of a k-tree and H/e is not.

Since $H/e \subseteq T/e$ we can instead look at the contraction of an edge in a k-tree rather than a contraction in a subset. Suppose we pick T as the smallest k-tree with a contraction such that T/e is not a subset of a k-tree

For starters if $T = K_{k+1}$ this is a contradiction because $K_{k+1}/e = K_k$ and $K_k \subset K_{k+1}$

Suppose now we write T as $T' + v$ by the definition given. If $v \in e$, then $T/e = T'$ which is a subset of T so $v \notin e$. Since $v \notin e$, then $e \in E(T')$ meaning T'/e would not be a subset of a k-tree. This creates a contradiction because we said that T was the smallest k-tree that has an edge contraction that is not a subset of a k-tree.